

Smart-MDT Reproducible Research Evaluation

Automatically generated evaluation framework

1 Overview

The evaluation contains 7360 benchmark rows, 46 datasets, and 8 methods from `rust_results_final_freeze_r10_1fbfa30`. Repeated runs and depths are averaged within each dataset, and paired comparisons use datasets as independent blocks. Bootstrap intervals use 10000 resamples, seed 20260718, and a 95.0% confidence level.

2 Executive findings

- Best mean accuracy: CALS-MDT (0.864820).
- Smallest mean tree: CompactExplain (11.124 nodes).
- Fastest mean fit time: Unary (0.047 seconds).
- Best mean AXp length: CompactExplain (2.721).
- For CALS-MDT vs CompactExplain on Accuracy, the right-hand method was lower by 0.0047 (95% bootstrap CI [-0.0073, -0.0022], raw p=0.001873, Holm-adjusted p=0.01124, paired Cohen's d=-0.523, medium effect).
- For SmartCertified vs CALS-MDT on Accuracy, the right-hand method was higher by 0.0052 (95% bootstrap CI [0.0014, 0.0092], raw p=0.005784, Holm-adjusted p=0.02892, paired Cohen's d=0.385, small effect).
- For CALS-MDT vs CompactExplain on Fit time (seconds), the right-hand method was lower by 6.0334 (95% bootstrap CI [-7.8811, -4.3309], raw p=2.842e-14, Holm-adjusted p=4.263e-13, paired Cohen's d=-0.965, large effect).
- For SmartCertified vs CALS-MDT on Fit time (seconds), the right-hand method was higher by 8.1640 (95% bootstrap CI [5.6094, 11.0623], raw p=5.446e-10, Holm-adjusted p=7.624e-09, paired Cohen's d=0.845, large effect).
- For SmartCertified vs CALS-MDT on Tree nodes, the right-hand method was lower by 37.9587 (95% bootstrap CI [-45.8178, -30.6804], raw p=3.522e-09, Holm-adjusted p=4.576e-08, paired Cohen's d=-1.411, large effect).
- For SmartCertified vs CompactExplain on Tree nodes, the right-hand method was lower by 39.5652 (95% bootstrap CI [-49.0461, -30.9891], raw p=3.52e-09, Holm-adjusted p=4.576e-08, paired Cohen's d=-1.254, large effect).

3 Descriptive statistics

Table 1: Central tendency and dispersion by method and metric.

Method	Metric	N	Mean	Median	Variance	Std.
Unary	Accuracy	46	0.859	0.901	0.014	0.118
Unary	Tree nodes	46	53.235	44.050	983.710	31.364
Unary	Predicate literals	46	26.117	21.525	245.927	15.682
Unary	Mean AXp length	46	3.927	3.916	1.435	1.198
Unary	Fit time (seconds)	46	0.047	0.012	0.009	0.094
Horn	Accuracy	46	0.860	0.903	0.014	0.119
Horn	Tree nodes	46	51.217	41.050	1061.101	32.575
Horn	Predicate literals	46	50.217	40.050	1061.101	32.575
Horn	Mean AXp length	46	5.137	5.110	2.281	1.510
Horn	Fit time (seconds)	46	0.065	0.017	0.018	0.135
AntiHorn	Accuracy	46	0.860	0.896	0.014	0.118
AntiHorn	Tree nodes	46	51.487	39.700	1089.997	33.015
AntiHorn	Predicate literals	46	50.487	38.700	1089.997	33.015
AntiHorn	Mean AXp length	46	5.553	5.115	4.229	2.057
AntiHorn	Fit time (seconds)	46	0.066	0.017	0.020	0.142
Square2CNF	Accuracy	46	0.859	0.890	0.014	0.120
Square2CNF	Tree nodes	46	50.807	40.450	1112.861	33.360
Square2CNF	Predicate literals	46	99.613	78.900	4451.442	66.719
Square2CNF	Mean AXp length	46	6.357	6.122	4.748	2.179
Square2CNF	Fit time (seconds)	46	0.444	0.073	1.043	1.021
Boolean Affine/GF(2)	Accuracy	46	0.854	0.881	0.014	0.119
Boolean Affine/GF(2)	Tree nodes	46	53.126	47.300	1060.428	32.564
Boolean Affine/GF(2)	Predicate literals	46	60.042	53.700	1238.913	35.198
Boolean Affine/GF(2)	Mean AXp length	46	7.897	8.185	5.437	2.332
Boolean Affine/GF(2)	Fit time (seconds)	46	0.162	0.031	0.120	0.346
SmartCertified	Accuracy	46	0.860	0.895	0.014	0.119
SmartCertified	Tree nodes	46	50.689	42.300	959.944	30.983
SmartCertified	Predicate literals	46	30.123	26.075	294.437	17.159
SmartCertified	Mean AXp length	46	4.184	4.195	1.335	1.156
SmartCertified	Fit time (seconds)	46	0.731	0.113	2.689	1.640
CALS-MDT	Accuracy	46	0.865	0.896	0.014	0.118
CALS-MDT	Tree nodes	46	12.730	12.800	70.748	8.411
CALS-MDT	Predicate literals	46	14.936	13.600	123.817	11.127
CALS-MDT	Mean AXp length	46	2.976	2.809	2.454	1.567
CALS-MDT	Fit time (seconds)	46	8.895	5.528	95.815	9.789
CompactExplain	Accuracy	46	0.860	0.895	0.014	0.118
CompactExplain	Tree nodes	46	11.124	9.850	51.889	7.203
CompactExplain	Predicate literals	46	12.786	9.550	125.144	11.187
CompactExplain	Mean AXp length	46	2.721	2.553	2.867	1.693
CompactExplain	Fit time (seconds)	46	2.861	1.752	15.606	3.950

Table 2: Ranges and 95 percent mean confidence intervals.

Method	Metric	Min.	Max.	95% CI lower	95% CI upper
Unary	Accuracy	0.543	1.000	0.824	0.894
Unary	Tree nodes	11.400	142.500	43.921	62.549
Unary	Predicate literals	5.200	70.750	21.460	30.774
Unary	Mean AXp length	1.500	6.759	3.571	4.282
Unary	Fit time (seconds)	0.001	0.524	0.019	0.075
Horn	Accuracy	0.536	1.000	0.824	0.895
Horn	Tree nodes	11.400	145.700	41.544	60.891
Horn	Predicate literals	10.400	144.700	40.544	59.891
Horn	Mean AXp length	1.594	8.976	4.689	5.586
Horn	Fit time (seconds)	0.002	0.739	0.025	0.105

Method	Metric	Min.	Max.	95% CI lower	95% CI upper
AntiHorn	Accuracy	0.528	1.000	0.825	0.895
AntiHorn	Tree nodes	11.400	138.800	41.683	61.291
AntiHorn	Predicate literals	10.400	137.800	40.683	60.291
AntiHorn	Mean AXp length	1.764	10.648	4.943	6.164
AntiHorn	Fit time (seconds)	0.002	0.802	0.024	0.109
Square2CNF	Accuracy	0.535	1.000	0.824	0.895
Square2CNF	Tree nodes	11.400	138.600	40.900	60.713
Square2CNF	Predicate literals	20.800	275.200	79.800	119.426
Square2CNF	Mean AXp length	2.847	11.357	5.710	7.004
Square2CNF	Fit time (seconds)	0.008	4.988	0.141	0.748
Boolean Affine/GF(2)	Accuracy	0.537	0.997	0.818	0.889
Boolean Affine/GF(2)	Tree nodes	11.400	144.900	43.456	62.796
Boolean Affine/GF(2)	Predicate literals	12.450	157.600	49.590	70.495
Boolean Affine/GF(2)	Mean AXp length	3.516	13.809	7.205	8.590
Boolean Affine/GF(2)	Fit time (seconds)	0.005	1.754	0.060	0.265
SmartCertified	Accuracy	0.544	1.000	0.824	0.895
SmartCertified	Tree nodes	11.400	141.500	41.488	59.890
SmartCertified	Predicate literals	5.200	76.750	25.027	35.218
SmartCertified	Mean AXp length	1.920	7.459	3.841	4.527
SmartCertified	Fit time (seconds)	0.008	8.272	0.244	1.218
CALS-MDT	Accuracy	0.537	0.999	0.830	0.900
CALS-MDT	Tree nodes	1.000	42.400	10.233	15.228
CALS-MDT	Predicate literals	0.000	51.050	11.631	18.240
CALS-MDT	Mean AXp length	0.000	8.273	2.511	3.442
CALS-MDT	Fit time (seconds)	0.340	44.199	5.988	11.801
CompactExplain	Accuracy	0.539	1.000	0.825	0.895
CompactExplain	Tree nodes	1.000	32.500	8.985	13.263
CompactExplain	Predicate literals	0.000	58.950	9.464	16.108
CompactExplain	Mean AXp length	0.000	9.274	2.219	3.224
CompactExplain	Fit time (seconds)	0.121	19.187	1.688	4.034

4 Pairwise significance

Positive differences and effect sizes indicate that the right-hand method has the larger metric value. Accuracy is maximized; nodes, literals, AXp length, and runtime are minimized. Holm-adjusted p-values control family-wise error across the configured tests.

Table 3: Paired Wilcoxon tests with Holm-adjusted p-values and bootstrap confidence intervals.

Comparison	Metric	Pairs	Mean diff.	Wilcoxon W	p-value	Holm p-value	Bootstrap lower	Bootstrap upper
SmartCertified vs CALS-MDT	Accuracy	46	0.005	273.000	0.006	0.029	0.001	0.009
SmartCertified vs CALS-MDT	Tree nodes	46	-37.959	0.000	0.000	0.000	-45.818	-30.680
SmartCertified vs CALS-MDT	Predicate literals	46	-15.187	72.000	0.000	0.000	-19.710	-10.872
SmartCertified vs CALS-MDT	Mean AXp length	46	-1.207	146.000	0.000	0.000	-1.635	-0.768
SmartCertified vs CALS-MDT	Fit time (seconds)	46	8.164	46.000	0.000	0.000	5.609	11.062
SmartCertified vs CompactExplain	Accuracy	46	0.001	525.000	0.871	0.871	-0.004	0.005
SmartCertified vs CompactExplain	Tree nodes	46	-39.565	0.000	0.000	0.000	-49.046	-30.989
SmartCertified vs CompactExplain	Predicate literals	46	-17.337	95.000	0.000	0.000	-23.148	-11.750
SmartCertified vs CompactExplain	Mean AXp length	46	-1.463	139.000	0.000	0.000	-1.962	-0.957
SmartCertified vs CompactExplain	Fit time (seconds)	46	2.131	122.000	0.000	0.000	1.132	3.300
CALS-MDT vs CompactExplain	Accuracy	46	-0.005	242.000	0.002	0.011	-0.007	-0.002
CALS-MDT vs CompactExplain	Tree nodes	46	-1.607	443.500	0.404	0.807	-3.511	0.102
CALS-MDT vs CompactExplain	Predicate literals	46	-2.150	382.500	0.128	0.383	-4.110	-0.317
CALS-MDT vs CompactExplain	Mean AXp length	46	-0.255	328.000	0.032	0.130	-0.543	0.057
CALS-MDT vs CompactExplain	Fit time (seconds)	46	-6.033	0.000	0.000	0.000	-7.881	-4.331

5 Dataset analysis

The Nemenyi critical difference for average accuracy ranks is 1.548.

Table 4: Cliff’s delta and paired Cohen’s d effect sizes.

Comparison	Metric	Cliff’s delta	Cliff magnitude	Cohen’s d	d magnitude
SmartCertified vs CALS-MDT	Accuracy	0.034	negligible	0.385	small
SmartCertified vs CALS-MDT	Tree nodes	-0.863	large	-1.411	large
SmartCertified vs CALS-MDT	Predicate literals	-0.552	large	-0.978	large
SmartCertified vs CALS-MDT	Mean AXp length	-0.491	large	-0.803	large
SmartCertified vs CALS-MDT	Fit time (seconds)	0.871	large	0.845	large
SmartCertified vs CompactExplain	Accuracy	0.000	negligible	0.041	negligible
SmartCertified vs CompactExplain	Tree nodes	-0.910	large	-1.254	large
SmartCertified vs CompactExplain	Predicate literals	-0.647	large	-0.866	large
SmartCertified vs CompactExplain	Mean AXp length	-0.617	large	-0.821	large
SmartCertified vs CompactExplain	Fit time (seconds)	0.698	large	0.568	medium
CALS-MDT vs CompactExplain	Accuracy	-0.032	negligible	-0.523	medium
CALS-MDT vs CompactExplain	Tree nodes	-0.105	negligible	-0.253	small
CALS-MDT vs CompactExplain	Predicate literals	-0.138	negligible	-0.328	small
CALS-MDT vs CompactExplain	Mean AXp length	-0.125	negligible	-0.243	small
CALS-MDT vs CompactExplain	Fit time (seconds)	-0.496	large	-0.965	large

Table 5: Best method and associated metrics for every dataset.

Dataset	Best method	Accuracy	Nodes	AXp	Runtime (s)
HTRU_2-bin	Unary	0.976	111.900	2.277	0.082
IndiansDiabetes-bin	CALS-MDT	0.741	14.100	3.871	22.738
Statlog_satellite-bin	CALS-MDT	0.960	24.100	4.166	3.862
adult_discretized	Boolean Affine/GF(2)	0.853	118.500	7.868	1.169
anneal	CALS-MDT	0.856	12.800	5.032	1.019
audiology	Horn	0.955	14.200	3.398	0.002
australian-credit	CompactExplain	0.864	5.800	1.358	1.582
balance-scale-bin	CALS-MDT	0.917	1.700	0.116	10.404
bank_conv-bin	Unary	0.902	93.100	6.442	0.071
banknote-bin	Square2CNF	0.990	30.200	5.054	0.052
biodeg-bin	CALS-MDT	0.818	20.300	5.128	21.699
breast-cancer-un	CALS-MDT	0.943	5.800	2.530	9.736
breast-wisconsin	CALS-MDT	0.950	7.000	2.704	7.035
car-un	Square2CNF	0.968	31.200	3.607	0.071
car_evaluation-bin	SmartCertified	0.942	38.400	2.585	0.014
compas_discretized	Unary	0.671	107.100	4.325	0.011
diabetes	Unary	0.744	83.500	5.342	0.012
forest-fires-un	SmartCertified	0.544	18.500	4.583	0.138
german-credit	CompactExplain	0.728	12.900	2.929	4.282
heart-cleveland	CALS-MDT	0.776	7.000	2.791	12.669
hepatitis	CALS-MDT	0.856	2.600	1.020	5.111
hypothyroid	Unary	0.980	42.300	1.932	0.011
ionosphere	CALS-MDT	0.878	5.900	2.174	2.077
kr-vs-kp	CompactExplain	0.988	28.500	4.228	4.464
letter	Square2CNF	0.989	72.900	6.225	1.480
letter_recognition-bin	AntiHorn	0.996	61.600	6.158	0.350
lymph	SmartCertified	0.819	21.700	3.675	0.011
magic04-bin	Square2CNF	0.837	138.600	7.620	3.381
messidor-bin	CompactExplain	0.643	9.400	2.548	0.162
mushroom	CompactExplain	1.000	9.800	4.938	1.877
pendigits	SmartCertified	0.996	43.700	3.727	0.704
primary-tumor	CompactExplain	0.815	11.100	1.840	1.785
segment	Boolean Affine/GF(2)	0.997	11.400	5.758	0.020
seismic.bumps-bin	CompactExplain	0.922	1.000	0.000	3.537
soybean	CALS-MDT	0.942	12.800	4.043	11.055
splice-1	Unary	0.955	74.300	4.291	0.058
taiwan_binarised	CALS-MDT	0.821	11.700	1.810	5.873

Dataset	Best method	Accuracy	Nodes	AXp	Runtime (s)
tic-tac-toe	Horn	0.918	47.200	4.726	0.006
titanic-un	CALS-MDT	0.819	16.300	2.969	8.384
vehicle	SmartCertified	0.946	36.600	3.956	0.092
vote	CompactExplain	0.945	6.200	1.923	1.516
wine1-un	AntiHorn	0.708	13.000	10.202	0.018
wine2-un	AntiHorn	0.664	15.700	10.425	0.018
wine3-un	CALS-MDT	0.751	3.800	4.053	0.359
winequality-red-bin	CompactExplain	0.994	1.000	0.000	0.379
yeast	CALS-MDT	0.717	21.400	3.888	2.067

Table 6: Dataset wins and average accuracy ranks. The critical difference uses a two-sided Nemenyi comparison at alpha 0.05.

Method	Wins	Average rank	Critical difference
CALS-MDT	15	3.120	1.548
Unary	6	4.130	1.548
SmartCertified	5	4.228	1.548
Horn	2	4.370	1.548
CompactExplain	9	4.522	1.548
Square2CNF	4	4.598	1.548
AntiHorn	3	4.685	1.548
Boolean Affine/GF(2)	2	6.348	1.548

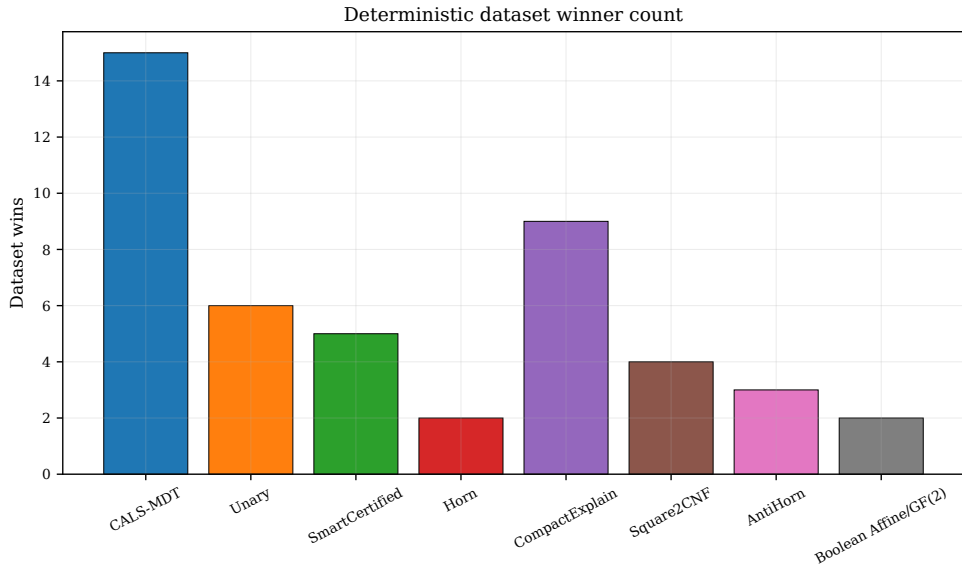


Figure 1: Dataset winner counts.

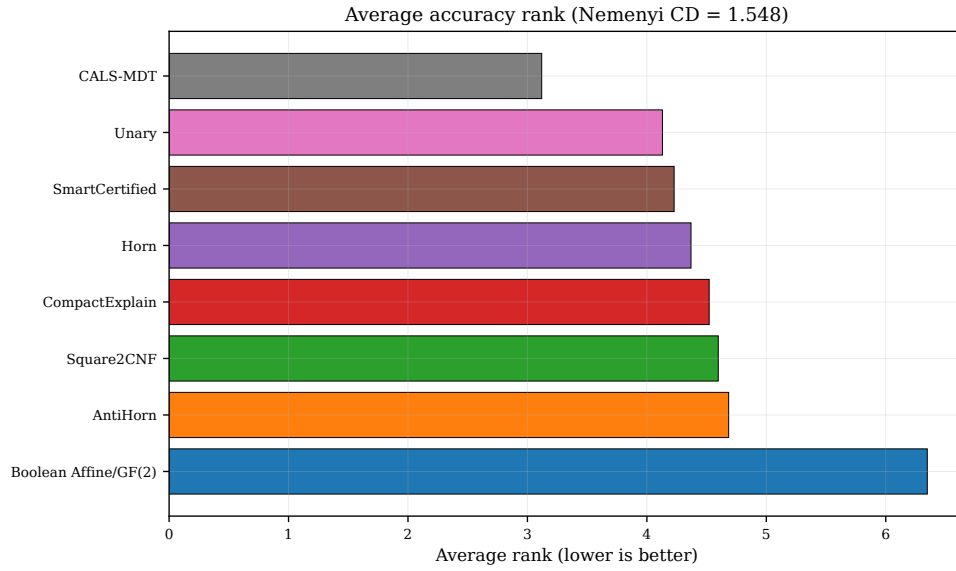


Figure 2: Average dataset accuracy ranks.

6 Performance and explanation figures

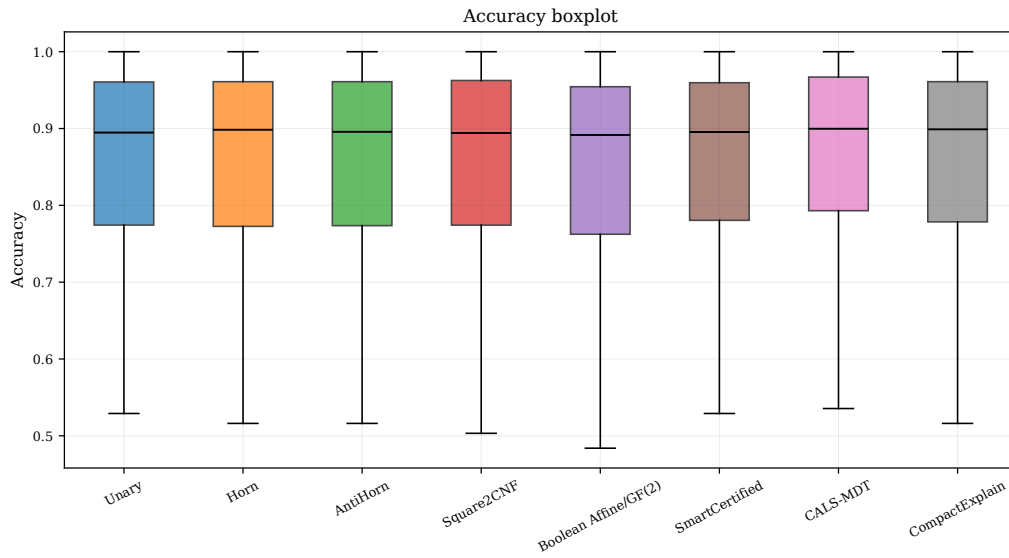


Figure 3: Accuracy boxplot.

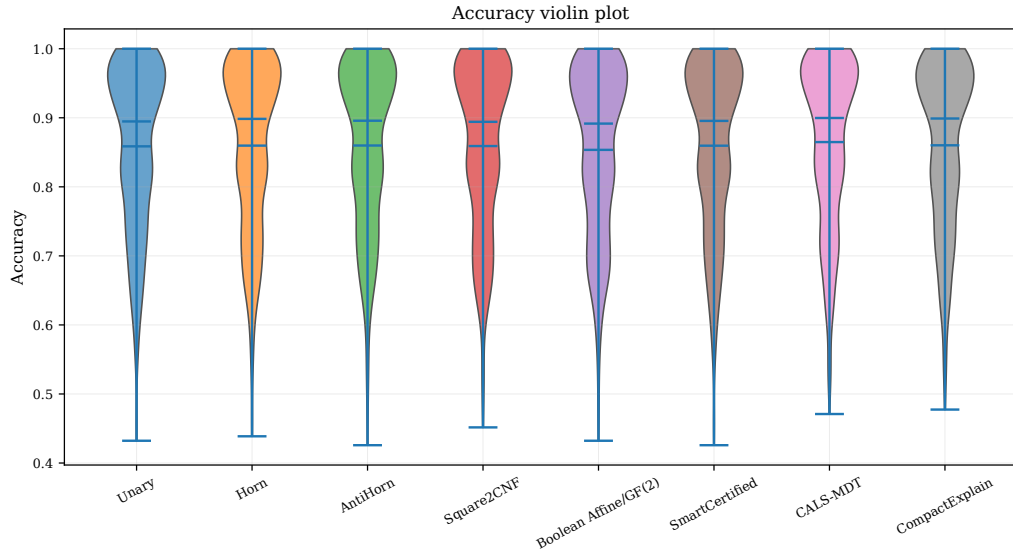


Figure 4: Accuracy violin plot.

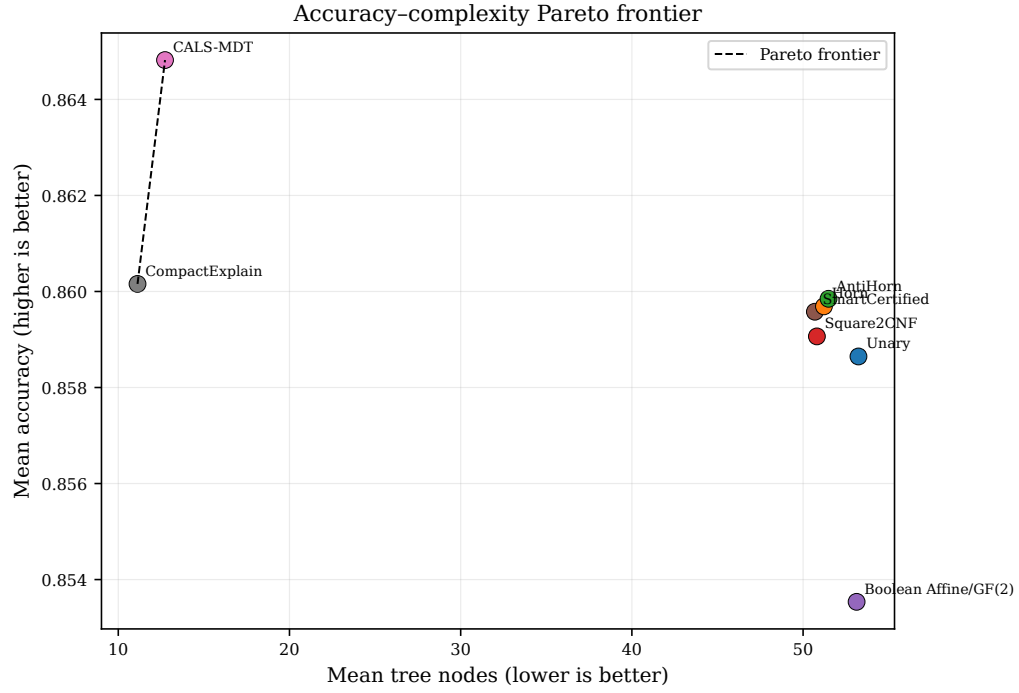


Figure 5: Accuracy-complexity Pareto frontier.

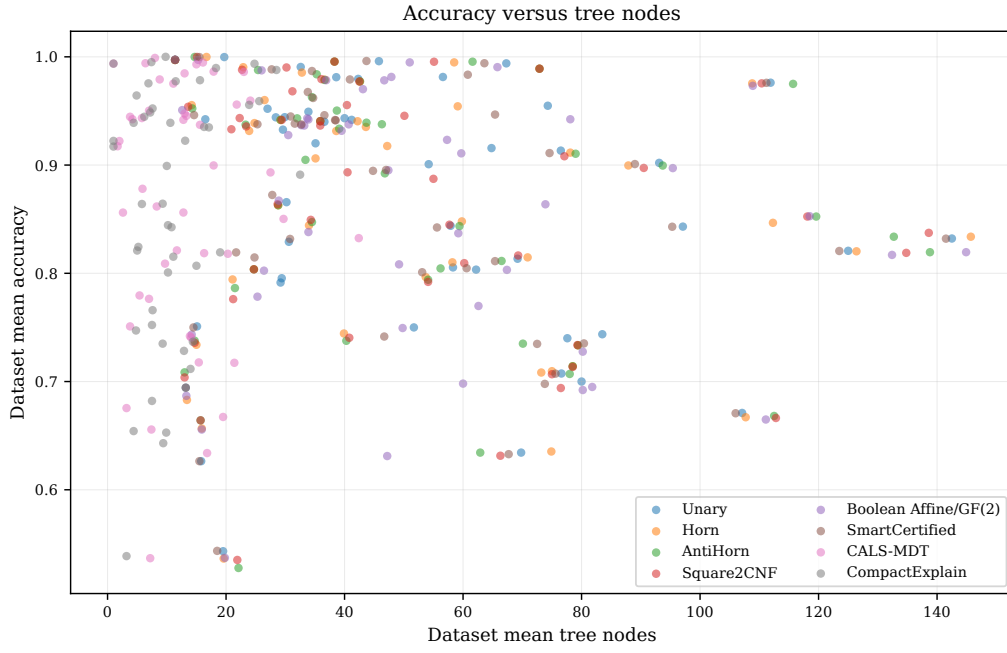


Figure 6: Accuracy versus tree nodes.

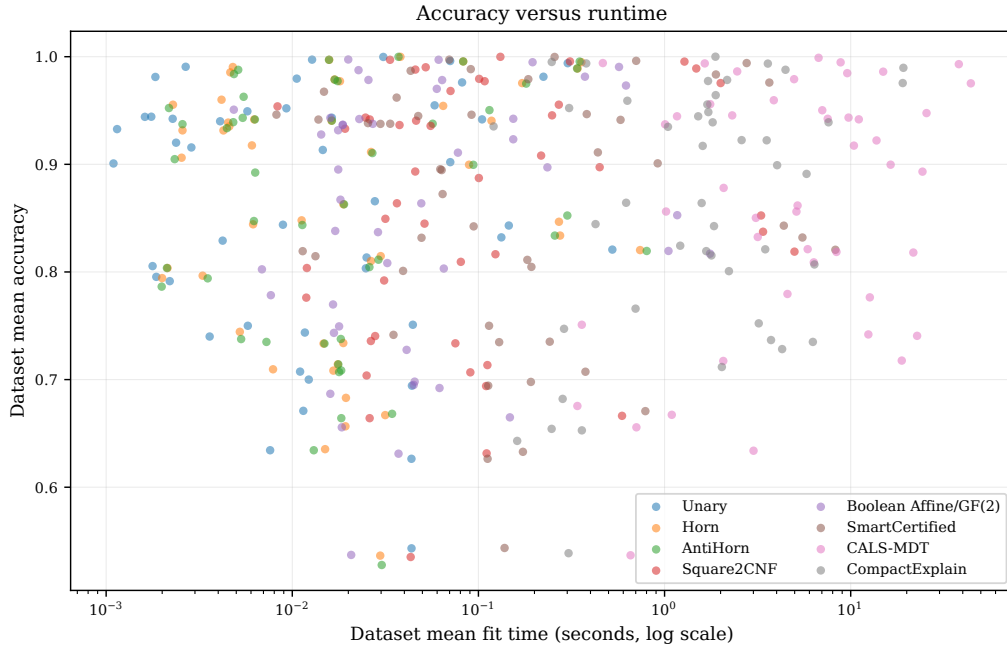


Figure 7: Accuracy versus runtime.

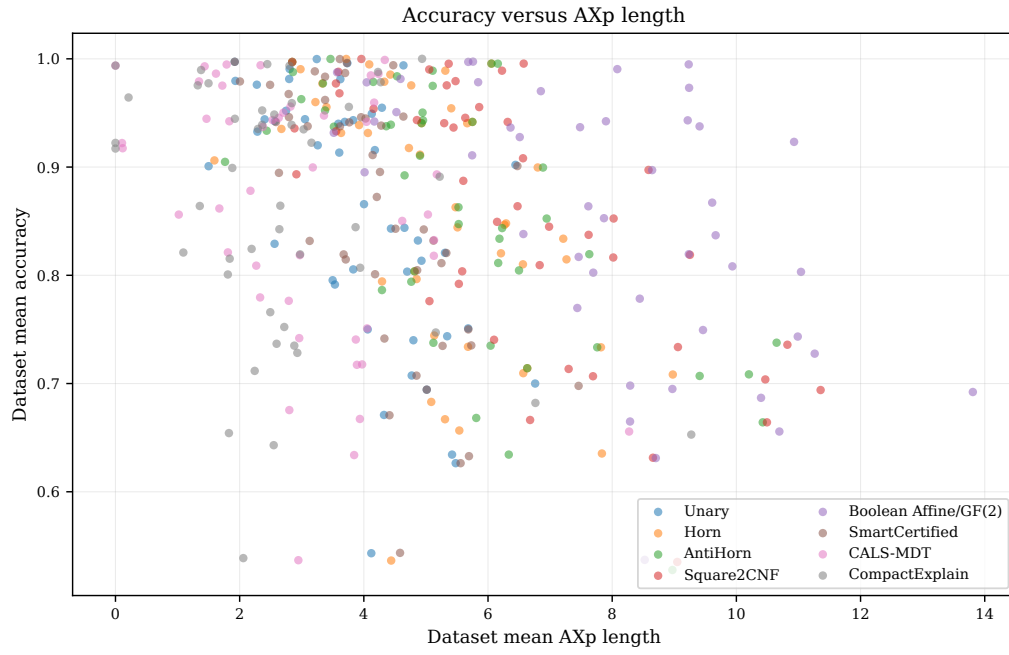


Figure 8: Accuracy versus AXp length.

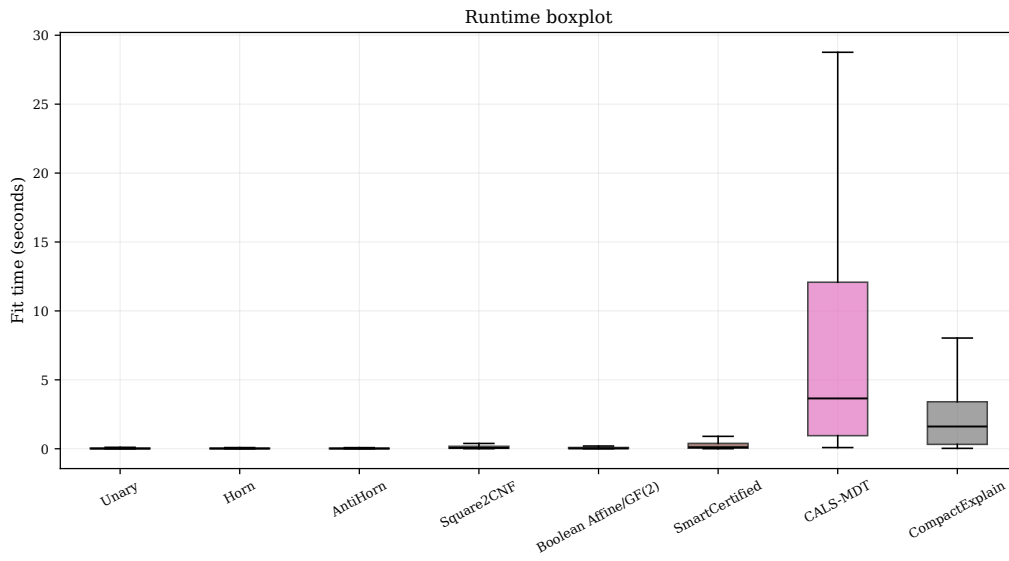


Figure 9: Runtime boxplot.

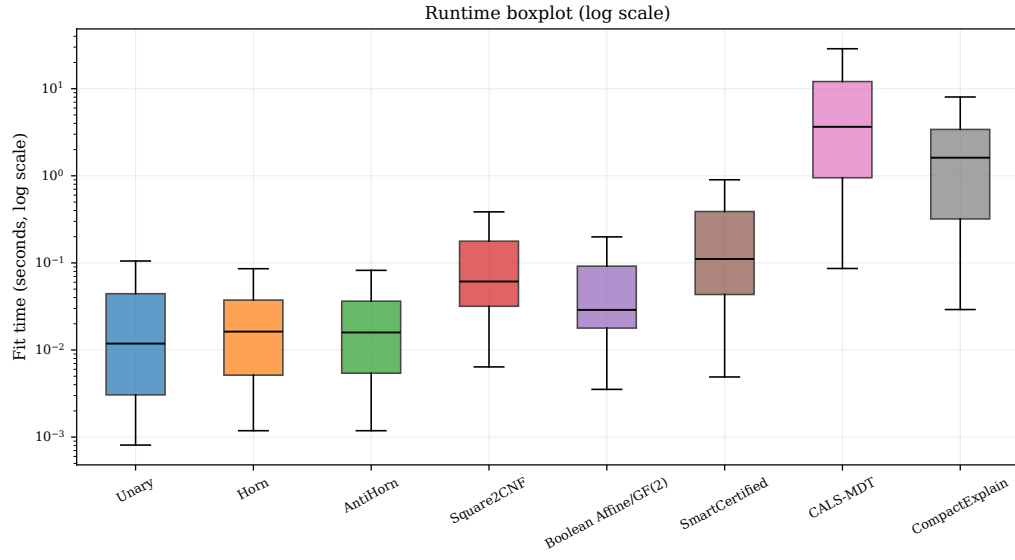


Figure 10: Runtime boxplot on a logarithmic scale.

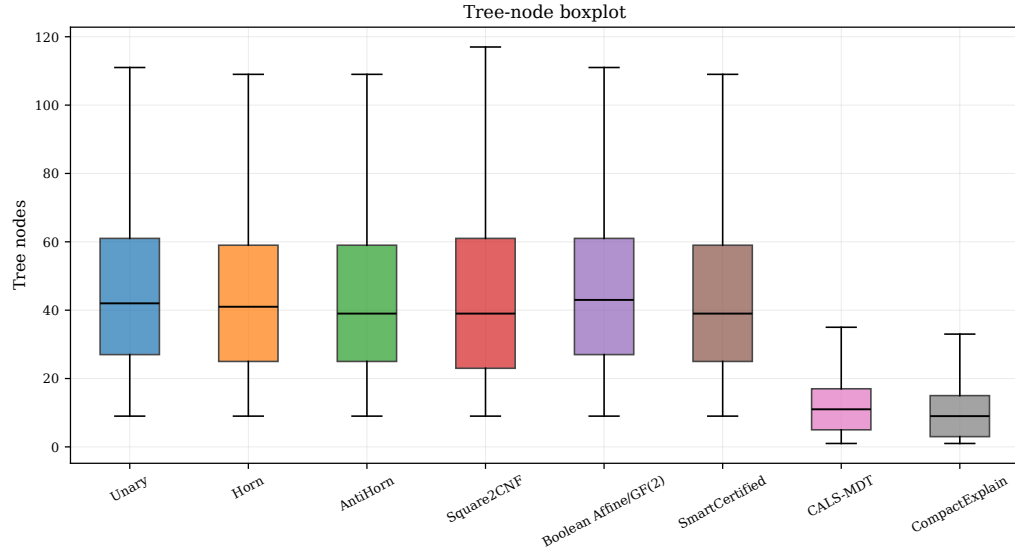


Figure 11: Tree-node boxplot.

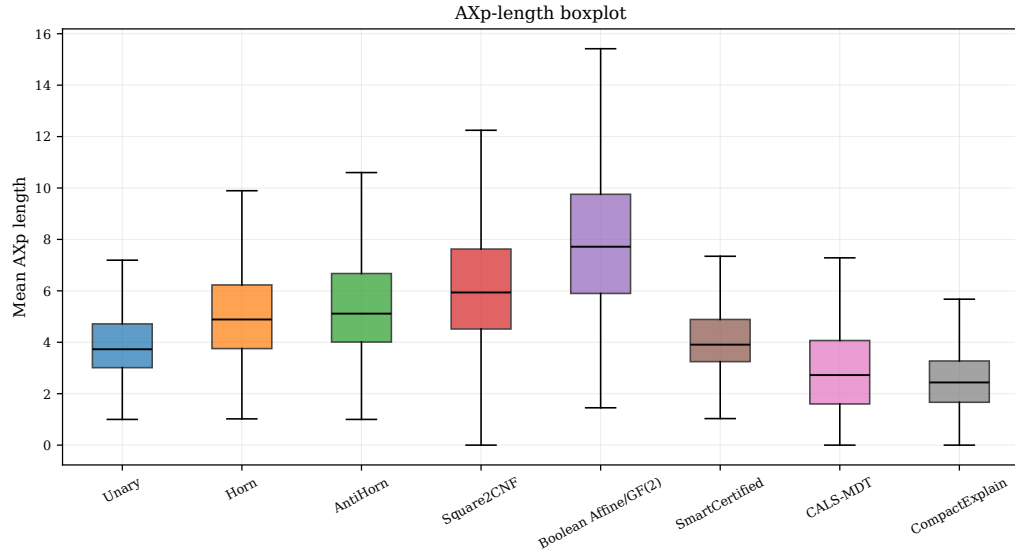


Figure 12: AXp-length boxplot.

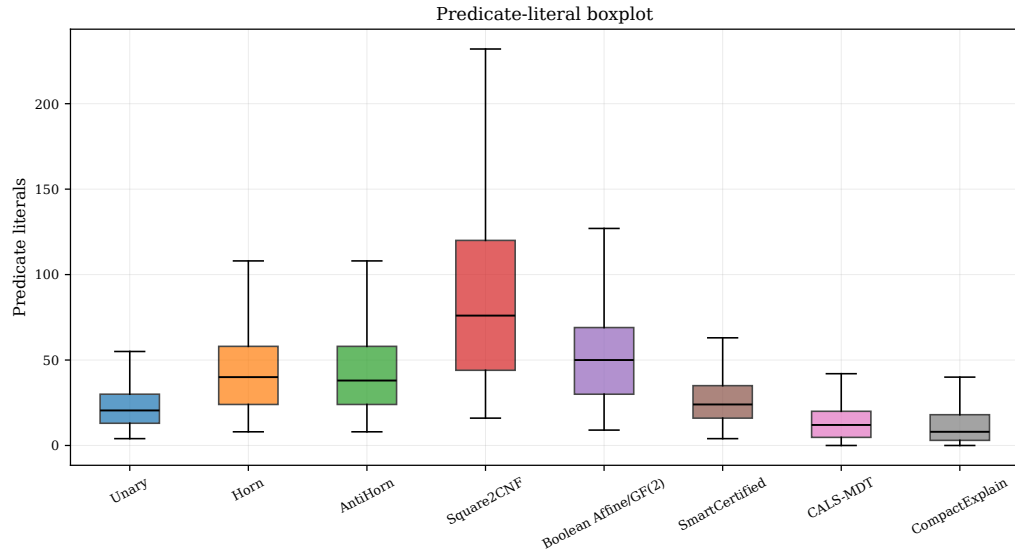


Figure 13: Predicate-literal boxplot.

Table 7: Certification and data-integrity audit.

Audit item	Count	Denominator	Percentage	evidence_available	status
datasets	46	46	100.000	1	pass
rows	7360	7360	100.000	1	pass
theorem_certified_rows	7360	7360	100.000	1	pass
empirical_rows	0	7360	0.000	1	pass
forbidden_predicates	0	7360	0.000	1	pass
feature_label_leakage	0	46	0.000	1	pass
path_violations	0	7360	0.000	1	pass
cached_subtree_violations	0	7360	0.000	1	pass
empirical_fallbacks	0	7360	0.000	1	pass
axp_evidence_violations	0	7360	0.000	1	pass
full_row_axp_violations	0	7360	0.000	1	pass
skipped_datasets	0	46	0.000	1	pass

7 Certification

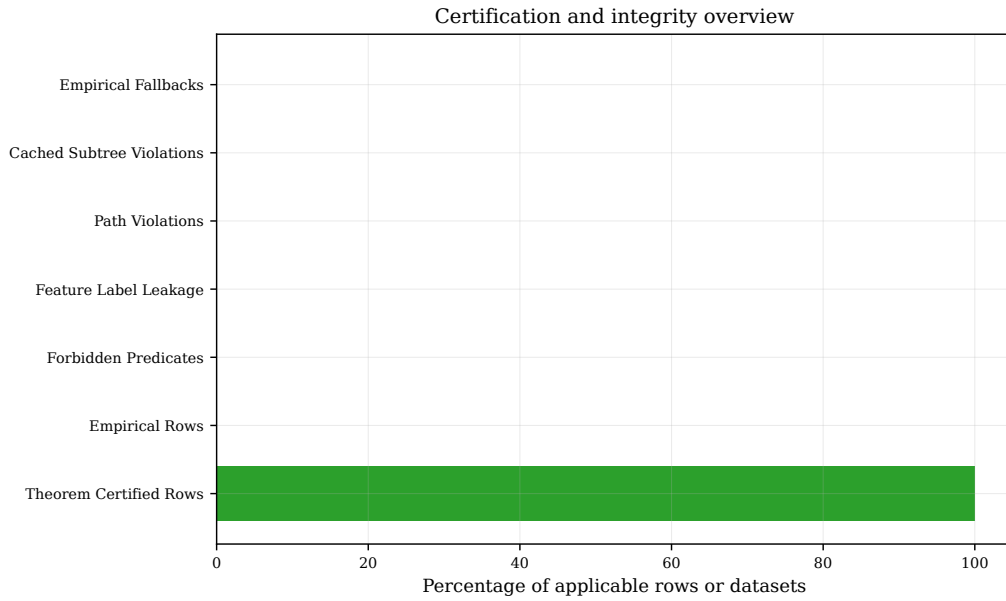


Figure 14: Certification and integrity audit.

8 Benchmark warnings

The warning summary contains 11 grouped records. The machine-readable table is available as `../tables/warnings_summary.csv`.

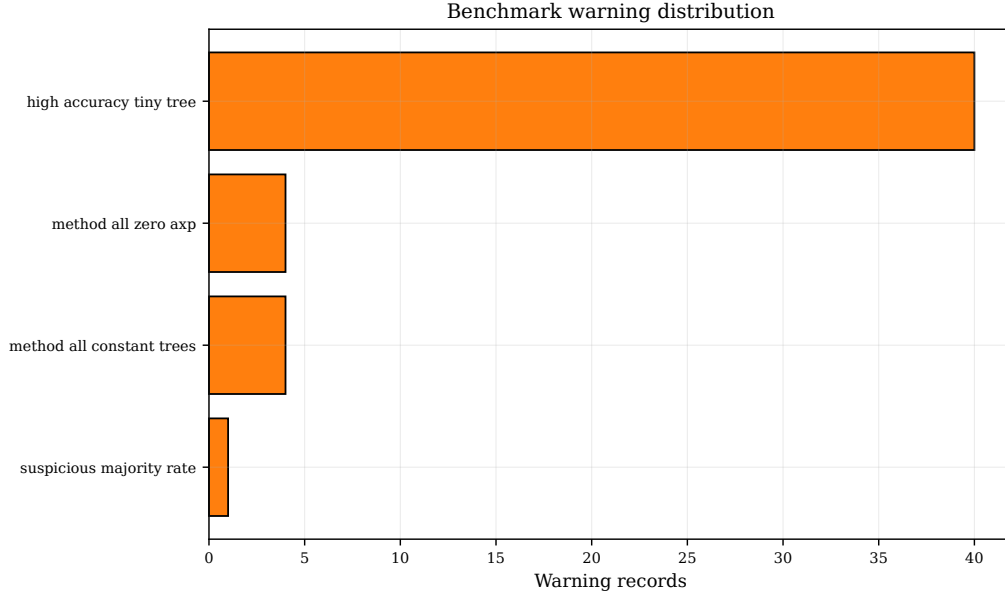


Figure 15: Benchmark warning distribution.

9 Search analysis

Table 8: Search-selection diagnostics.

Method	Rows	Greedy nodes	Lookahead nodes	B&B active	B&B avoided
Unary	920	0.000	0.000	0.000	0.000
Horn	920	0.000	0.000	0.000	0.000
AntiHorn	920	0.000	0.000	0.000	0.000
Square2CNF	920	0.000	0.000	0.000	0.000
Boolean Affine/GF(2)	920	0.000	0.000	0.000	0.000
SmartCertified	920	0.000	0.000	0.000	0.000
CALS-MDT	920	34304.000	0.000	29610.000	0.000
CompactExplain	920	24053.000	11604.000	0.000	27756.000

Table 9: Cache activation, candidate savings, and search time.

Method	Cache active	Candidate savings	Cache hits	Search time (s)
Unary	0.000	0.000	2256713.000	43.420
Horn	0.000	0.000	4417514.000	59.647
AntiHorn	0.000	0.000	4367932.000	61.040
Square2CNF	0.000	0.000	5111175.000	408.936
Boolean Affine/GF(2)	0.000	0.000	4906929.000	149.459
SmartCertified	0.000	0.000	18141420.000	672.113
CALS-MDT	55196.000	1584533.000	38342310.000	8182.759
CompactExplain	56805.000	458777.000	24730873.000	2632.005

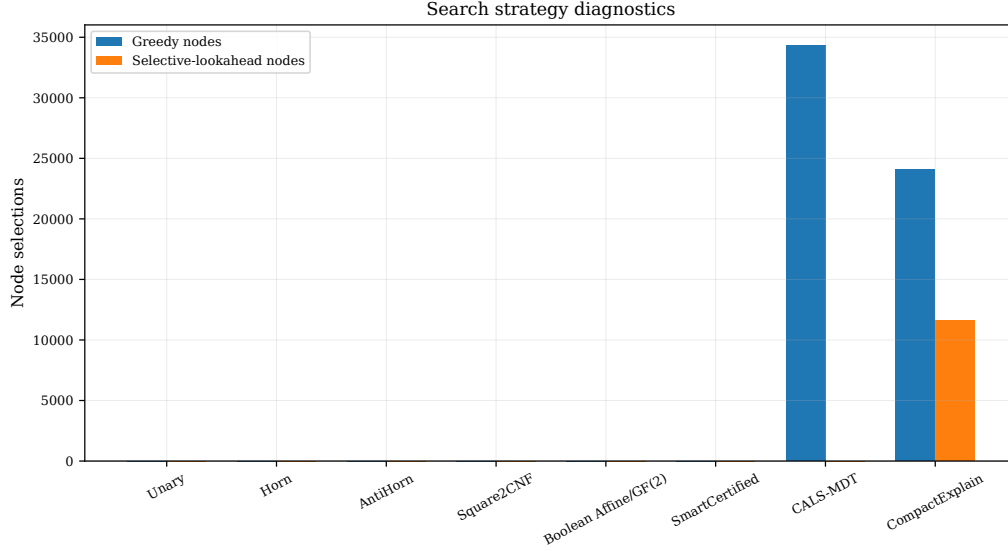


Figure 16: Greedy and selective-lookahead node selections.

10 Pruning analysis

Table 10: Validation metrics before and after pruning.

Method	Accuracy before	Accuracy after	Balanced before	Balanced after	Minority recall before	Minority recall after
Unary	0.000	0.000	0.000	0.000	0.000	0.000
Horn	0.000	0.000	0.000	0.000	0.000	0.000
AntiHorn	0.000	0.000	0.000	0.000	0.000	0.000
Square2CNF	0.000	0.000	0.000	0.000	0.000	0.000
Boolean Affine/GF(2)	0.000	0.000	0.000	0.000	0.000	0.000
SmartCertified	0.000	0.000	0.000	0.000	0.000	0.000
CALS-MDT	0.854	0.879	0.789	0.805	0.673	0.670
CompactExplain	0.854	0.869	0.789	0.786	0.672	0.637

Table 11: Tree-size reduction during pruning.

Method	Rows	Nodes before	Nodes after	Tree reduction (%)
Unary	920	53.235	53.235	0.000
Horn	920	51.217	51.217	0.000
AntiHorn	920	51.487	51.487	0.000
Square2CNF	920	50.807	50.807	0.000
Boolean Affine/GF(2)	920	53.126	53.126	0.000
SmartCertified	920	50.689	50.689	0.000
CALS-MDT	920	46.133	12.730	72.405
CompactExplain	920	46.659	11.124	76.159

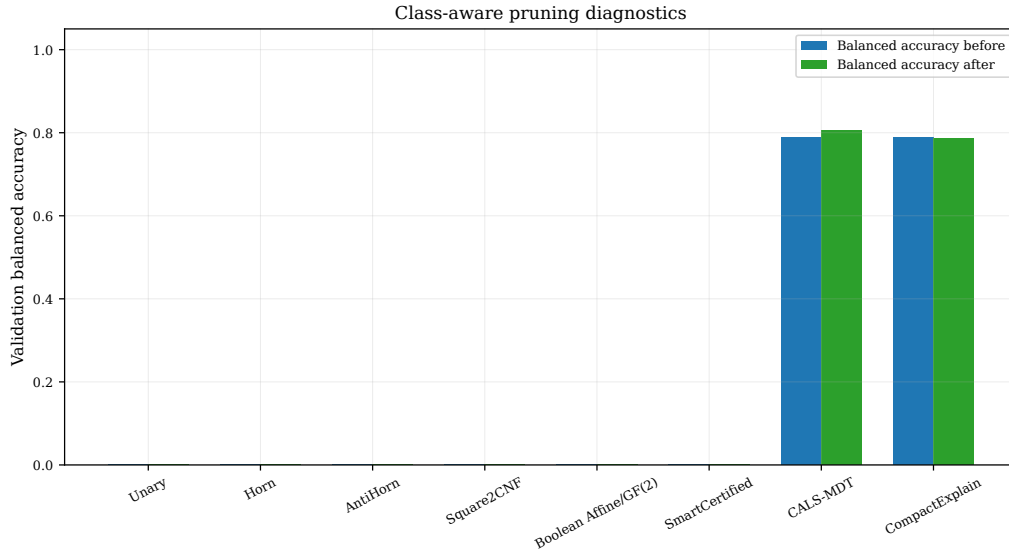


Figure 17: Validation balanced accuracy before and after pruning.

11 AXp analysis

Table 12: AXp distribution by method.

Method	Rows	Mean AXp	Median AXp	AXp std.	Min. AXp	Max. AXp
Unary	920	3.927	3.728	1.531	0.000	9.618
Horn	920	5.137	4.887	1.892	0.000	12.863
AntiHorn	920	5.553	5.114	2.389	0.000	14.523
Square2CNF	920	6.357	5.938	2.575	0.000	17.542
Boolean Affine/GF(2)	920	7.897	7.717	2.896	0.000	20.000
SmartCertified	920	4.184	3.907	1.531	0.000	10.673
CALS-MDT	920	2.976	2.724	2.135	0.000	14.887
CompactExplain	920	2.721	2.438	2.096	0.000	15.491

Table 13: Selected predicate-family usage.

Method	Family usage
Unary	Unary:24028
Horn	Horn:23100
AntiHorn	AntiHorn:23224
Square2CNF	Square2Cnf:22911
Boolean Affine/GF(2)	Affine:23978
SmartCertified	Affine:1121 — AntiHorn:682 — Horn:2348 — Square2Cnf:150 — Unary:18556
CALS-MDT	Affine:354 — AntiHorn:586 — Horn:1586 — Square2Cnf:1871 — Unary:999
CompactExplain	Affine:281 — AntiHorn:414 — Horn:1063 — Square2Cnf:1720 — Unary:1179

12 Discussion

Predictive performance, compactness, runtime, explanation length, and certification are reported separately. Paired Wilcoxon tests preserve the dataset blocks after averaging repeated runs and depths. Bootstrap intervals and effect sizes complement p-values with uncertainty and magnitude.

The generated reproducibility manifest records input hashes, package versions, seeds, and output files.